

Quantifying the Sum-Rate Cost of Reciprocity in Beyond-Diagonal RIS-Aided MU-MISO Systems: A Manifold Optimisation Comparison Across Single-, Group-, and Fully-Connected Architectures

ECE 310 Wireless Communications — Milestone 1 Report

Group 9 | Ahmedabad University

Aashi Bhimjiyani (AU2440027)

Kahaan Shah (AU2440047)

Prakshal Shah (AU2440265)

Samridhi Mehrotra (AU2440033)

September 13, 2026

Abstract

Beyond-diagonal reconfigurable intelligent surfaces (BD-RIS) extend conventional RIS by permitting non-diagonal scattering matrices, adding degrees of freedom for sum-rate maximisation in multi-user multiple-input single-output (MU-MISO) systems. Unconstrained unitary designs achieve higher theoretical sum rates but violate physical reciprocity and require active hardware. Reciprocal designs, which enforce a symmetric scattering matrix through manifold optimisation, can be implemented with passive components — though at some sum-rate cost. This report defines the research problem, reviews the relevant literature, and identifies the state-of-the-art position for this project. The core question — how much sum-rate is lost by enforcing reciprocity (the "reciprocity gap") across single-connected (CWSC), group-connected (CWGC), and fully-connected (CWFC) BD-RIS architectures — remains unanswered in the literature and is the primary contribution this project targets.

I. INTRODUCTION

Reconfigurable intelligent surfaces (RIS) have attracted considerable interest as a low-cost way to control wireless propagation. Conventional RIS use diagonal scattering matrices, meaning each element reflects signals independently. Beyond-diagonal RIS (BD-RIS), introduced by Li et al. [1], drops this restriction and allows non-diagonal scattering matrices, which create inter-element signal paths and open up additional optimisation degrees of freedom.

BD-RIS architectures differ in how elements are connected internally: single-connected (CWSC), where each element connects to one port; group-connected (CWGC), where elements are partitioned into internally-connected groups; and fully-connected (CWFC), where all elements are mutually coupled. Moving from CWSC to CWFC increases both circuit complexity and achievable performance.

A key physical consideration for passive BD-RIS is reciprocity. Under the passive, lossless reciprocal-network model, the scattering matrix must be symmetric, $\Theta = \Theta^T$ [3]. Unconstrained unitary designs do not impose this symmetry and therefore serve as an idealised performance benchmark rather than a complete model of a reciprocal passive implementation. Additional mechanisms may be required if a non-reciprocal response is desired.

Fidanovski et al. [2] address passive-realisation constraints by optimising over the symmetric unitary manifold using conjugate gradient ascent (CGA). However, the sum-rate difference between reciprocal and unconstrained designs is not the central quantity reported in that work.

This project addresses that comparison directly. Both designs will be evaluated under the same channel conditions and fixed transmit beamforming, and the sum-rate difference will be measured across CWSC, CWGC, and CWFC as a function of RIS size M , number of groups G , and transmit power $P_{\max}$

II. PROBLEM STATEMENT

A. System Model

We consider a MU-MISO downlink system with a base station (BS) of N antennas serving K single-antenna users through a BD-RIS of M passive elements. The received signal at user k is:

$$y_k = (h_{d,k}^H + h_{r,k}^H \Theta G) w_k s_k + \text{interference} + \text{noise}$$

where $h_{d,k} \in \mathbb{C}^N$ is the direct BS-user channel, $G \in \mathbb{C}^{M \times N}$ is the BS-RIS channel, $h_{r,k} \in \mathbb{C}^M$ is the RIS-user channel, $\Theta \in \mathbb{C}^{M \times M}$ is the BD-RIS scattering matrix, and $w_k \in \mathbb{C}^N$ is the beamforming vector for user k . The sum-rate is

$$R = \sum_k \log_2(1 + \text{SINR}_k)$$

B. Scattering Matrix Constraints

The scattering matrix is constrained to be unitary, $\Theta^H \Theta = I$, to represent a passive and lossless scattering model. In the unconstrained benchmark, Θ is optimised over the unitary group $U(M)$. In the reciprocal design, the additional symmetry condition $\Theta = \Theta^T$ is imposed, restricting the feasible set to the symmetric unitary subset $U_S(M) \subset U(M)$. The reciprocity gap is defined as the difference in sum-rate between these two cases under otherwise identical conditions.

The detailed manifold geometry, retractions, and optimisation updates will be developed during Milestone 2, when the simulation and optimisation framework is implemented.

C. Research Question

What is the sum-rate cost of enforcing a symmetric scattering matrix $\Theta = \Theta^T$ in a BD-RIS system, compared to an unconstrained unitary design, under the same channel realisations and fixed beamforming, across CWSC, CWGC, and CWFC architectures?

D. Scope and Assumptions

Scope: This project focuses on scattering matrix optimisation with fixed (pre-computed) transmit beamforming. Keeping the beamformer fixed isolates the effect of the reciprocity constraint cleanly; joint optimisation is out of scope.

Assumptions: Rayleigh and Rician small-scale fading with distance-dependent path loss ($\zeta_0 = -30$ dB at $d_0 = 1$ m, exponent $\varepsilon = 2.2$); BS-RIS distance ~ 50 m; RIS-user distance ~ 2.5 m; noise power $N_0 = -80$ dBm. Parameter ranges: $N = K \in \{2, \dots, 6\}$, $M \in \{8, \dots, 64\}$, $G \in \{1, 2, 4, M\}$, $P_{\max} \in [0, 20]$ dBm.

III. LITERATURE REVIEW

The literature falls into three groups: BD-RIS architecture and design, performance analysis and extensions, and the optimisation methods both designs rely on.

A. BD-RIS Foundations and Architecture

Before BD-RIS, the diagonal RIS model was standard — each element reflects independently, which caps the degrees of freedom available for optimisation.

Li et al. [1] introduced the unified BD-RIS framework covering CWSC, CWGC, and CWFC architectures. They optimise the scattering matrix as a general unitary matrix over the Stiefel manifold, without any symmetry constraint. The unconstrained design achieves the highest sum-rate within each architecture class, but it is not physically realisable with passive-only components. The paper also establishes the performance ordering: $CWFC > CWGC > CWSC$, at increasing circuit complexity.

Shen et al. [3] give the physical justification for the reciprocity constraint. Through scattering parameter network analysis, they show that a passive lossless RIS — built from reconfigurable impedance elements such as varactors or PIN diodes — must satisfy $\Theta = \Theta^T$ by the Lorentz reciprocity theorem. Breaking this symmetry requires active components, which increase power and cost. This result is why reciprocity is a necessary condition for any purely passive BD-RIS.

Fidanovski et al. [2] tackle passive implementation by designing the BD-RIS scattering matrix on the symmetric unitary manifold. They use fractional programming (Dinkelbach/quadratic transform) as the outer loop and conjugate gradient ascent (CGA) on the symmetric Stiefel manifold — via Manopt — as the inner solver, under fixed transmit beamforming. The result is a passive-realisable design. However, they do not compare against the unconstrained case under the same conditions, so the sum-rate loss from enforcing symmetry is not reported.

B. BD-RIS Performance Analysis and Extensions

Yahya et al. [4] derive closed-form passive maximum ratio transmission (MRT) and interference nulling conditions for BD-RIS. Their analysis shows that the off-diagonal elements in BD-RIS enable interference suppression that diagonal RIS cannot achieve. Although we use numerical optimisation rather than closed-form design, this paper gives theoretical grounding for why CWFC outperforms the other architectures and how much headroom exists with non-diagonal scattering.

Wu and Clerckx [5] propose a universal fractional programming and manifold optimisation framework that covers arbitrary BD-RIS architectures, including hybrid and partially-connected variants beyond the standard three. They parameterise all cases under a common block-diagonal structure indexed by group size. This is useful background for our implementation, since it clarifies the structural relationship between CWSC, CWGC, and CWFC — though the reciprocity constraint is not addressed.

Nerini and Clerckx [6] map out the Pareto frontier between sum-rate and circuit complexity for BD-RIS. Using tunable impedance element count as the complexity metric, they show CWFC requires $O(M^2)$ elements versus $O(M)$ for CWSC, with CWGC between them. Our project extends this Pareto picture: the reciprocity constraint reduces the complexity of any architecture (since passive circuits are simpler than active ones), so enforcing symmetry shifts each architecture's position on the frontier — by how much is part of what we aim to quantify.

C. Optimisation Tools

Shen and Yu [7] develop the fractional programming framework used in both [1] and [2]. Their quadratic transform converts the sum-rate objective — a sum of $\log(1 + \text{SINR})$ terms — into a tractable auxiliary problem that decouples across users. We will adopt the same FP reformulation as the outer loop in our simulation, which ensures a fair comparison: any sum-rate difference between the two inner solvers reflects the reciprocity constraint alone, not differences in the outer optimisation.

Boumal [8] provides the mathematical background for optimisation on Riemannian manifolds, covering both the Stiefel manifold (full unitary group $U(M)$) and symmetric submanifolds. The CGA algorithm in [2] follows directly from this framework. Understanding the geometry of $U(M)$ versus $U_S(M)$ matters for implementing both solvers and for interpreting convergence behaviour when the two manifolds diverge at larger M .

D. Summary Table

Table I summarises the key attributes of the primary references:

Ref	Paper	Key Contribution	Architecture / Constraint	Gap / Limitation
[1]	Li et al., IEEE TWC 2023	Unified BD-RIS model (CWSC/CWGC/CWFC); unitary scattering matrix via Stiefel manifold	Unconstrained ($\Theta \in U(M)$); all three architectures	No reciprocity enforcement; not realisable with passive-only hardware
[2]	Fidanovski et al., IEEE TWC 2026	Reciprocal BD-RIS via CGA on symmetric unitary manifold; passive hardware enabled	Symmetric ($\Theta = \Theta^T$); fixed beamforming	Reciprocity gap vs. unconstrained not quantified

Ref	Paper	Key Contribution	Architecture / Constraint	Gap / Limitation
[3]	Shen et al., IEEE TWC 2022	Physical basis for reciprocity in RIS via scattering parameter analysis	Passive/lossless impedance networks	Does not address BD-RIS optimisation
[4]	Yahya et al., IEEE OJ-COMS 2024	Passive MRT and interference nulling with BD-RIS	Closed-form analysis; no constraint comparison	No simulation of reciprocal vs. unconstrained gap
[5]	Wu & Clerckx, arXiv 2024	Universal FP + manifold framework for arbitrary BD-RIS architectures	General block-diagonal structure	Reciprocity constraint not addressed
[6]	Nerini & Clerckx, IEEE CL 2023	Pareto frontier: performance vs. circuit complexity for BD-RIS	Complexity metrics per architecture	Reciprocity as a complexity factor not included
[7]	Shen & Yu, IEEE TSP 2018	Fractional programming (quadratic transform) for SINR-based objectives	General FP framework	Not specific to BD-RIS or manifold constraints
[8]	Boumal, Cambridge 2023	Riemannian optimisation on smooth manifolds (Stiefel, symmetric submanifolds)	Mathematical framework	Textbook — no wireless application

IV. STATE-OF-THE-ART POSITION

A. Current SOTA

The state of the art splits cleanly along the reciprocity axis. For sum-rate performance, Li et al. [1] set the benchmark: their CWFC design gives the highest achievable sum-rate under a unitary constraint without symmetry, and Wu and Clerckx [5] extend this to a broader class of architectures without changing that position. For passive-realisable designs, Fidanovski et al. [2] are the current state of the art: their CGA-based manifold optimisation gives the best known sum-rate under the symmetry constraint.

B. The Gap This Project Addresses

No published work compares [1] and [2] under controlled, identical conditions. Four specific gaps follow from this:

(1) Same channel: Neither paper uses the same channel realisations, path loss model, and noise parameters to evaluate both designs side-by-side.

(2) Same beamforming: The two papers couple scattering matrix design with beamforming differently. Fixing the beamformer lets us attribute any sum-rate difference to the scattering matrix constraint alone.

(3) Quantified gap: No paper reports the reciprocity gap in bps/Hz or % as a function of M , G , or $P_{\max}$. This is the main output this project will produce.

(4) Pareto extension: Nerini and Clerckx [6] characterise the complexity-performance frontier for unconstrained designs only. Adding the reciprocity constraint as a design axis extends that picture.

C. Positioning Statement

This project sits between [1] and [2]. We take the algorithms from both, run them on the same channel instances with a fixed beamformer, and measure the sum-rate gap that comes from enforcing symmetry. The outcome will be the first controlled, cross-architecture characterisation of the reciprocity gap in BD-RIS.

V. PROJECT TIMELINE

Table II lists the planned milestones across the project semester:

Milestone	Focus	Key Deliverables
M1	Foundation	Problem definition, literature review, SOTA positioning, initial repo structure, channel model skeleton
M2	Implementation	Full simulation environment; reproduction of baseline curves from [1] and [2]; fixed beamformer implementation; channel generation (Rayleigh + Rician)
M3	Analysis	Reciprocity gap measurement across all architectures; parametric sweeps over M , G , $P_{\max}$; Pareto chart (complexity vs. sum-rate); convergence analysis
M4	Reporting	IEEE-format final report; clean documented GitHub repo; presentation slides; video walkthrough

VI. CONCLUSION

This report lays out the technical groundwork for the ECE 310 BD-RIS project. The problem is well-defined: measure the sum-rate cost of enforcing a symmetric scattering matrix across CWSC, CWGC, and CWFC architectures. The literature review covers eight references spanning BD-RIS design, performance analysis, and the optimisation tools both algorithms use. The SOTA survey confirms that the reciprocity gap has not been directly measured under controlled conditions in prior work — that is the contribution this project will make. Four milestones are planned, ending with a reproducible simulation codebase and an IEEE-format final report.

REFERENCES

- [1] H. Li, S. Shen, and B. Clerckx, "Beyond Diagonal Reconfigurable Intelligent Surfaces: From Transmitting and Reflecting Modes to Single-, Group-, and Fully-Connected Architectures," *IEEE Transactions on Wireless Communications*, vol. 22, no. 4, pp. 2311–2324, Apr. 2023.

- [2]M. Fidanovski, I. A. Morales Sandoval, H. S. Rou, G. T. F. de Abreu, and E. Björnson, "Reciprocal Beyond-Diagonal Reconfigurable Intelligent Surface (BD-RIS): Scattering Matrix Design via Manifold Optimisation," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 20463–20477, 2026.
- [3]S. Shen, B. Clerckx, and R. Murch, "Modelling and Architecture Design of Reconfigurable Intelligent Surfaces Using Scattering Parameter Network Analysis," *IEEE Transactions on Wireless Communications*, vol. 21, no. 2, pp. 1229–1243, Feb. 2022.
- [4]H. Yahya, H. Li, M. Nerini, B. Clerckx, and M. Debbah, "Beyond Diagonal RIS: Passive Maximum Ratio Transmission and Interference Nulling Enabler," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 7613–7627, 2024.
- [5]Z. Wu and B. Clerckx, "Optimisation of Beyond Diagonal RIS: A Universal Framework Applicable to Arbitrary Architectures," *arXiv:2412.15965*, 2024.
- [6]M. Nerini and B. Clerckx, "Pareto Frontier for the Performance-Complexity Trade-Off in Beyond Diagonal Reconfigurable Intelligent Surfaces," *IEEE Communications Letters*, vol. 27, no. 10, pp. 2842–2846, Oct. 2023.
- [7]K. Shen and W. Yu, "Fractional Programming for Communication Systems—Part I: Power Control and Beamforming," *IEEE Transactions on Signal Processing*, vol. 66, no. 10, pp. 2616–2630, May 2018.
- [8]N. Boumal, *An Introduction to Optimization on Smooth Manifolds*. Cambridge, U.K.: Cambridge University Press, 2023.